

Hallucinating our way to AI-backed documentation

PART I - THEORY

Who am I

Who am I

- AI Solutions Architect at Cennso

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- AI Solutions Architect at Cennso
- Independent open-source maintainer - brainfart.dev

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Lukasz Gornicki

Pushing the boundaries in AsyncAPI
Initiative and Open Source



The big words

The big words



The big words

 **Agentic**

 **LLM**

The big words

 **Agentic**

 **LLM**

 **RAG**

The big words

 **Agentic**

 **LLM**

 **RAG**

 **MCP**

The big words

 **Agentic**

 **LLM**

 **RAG**

 **MCP**

 **Skills**

What is Agentic AI

What is Agentic AI

AI without copy/paste

What is Agentic AI

AI without copy/paste

Chat AI:

What's on the agenda today?

+ Ask anything



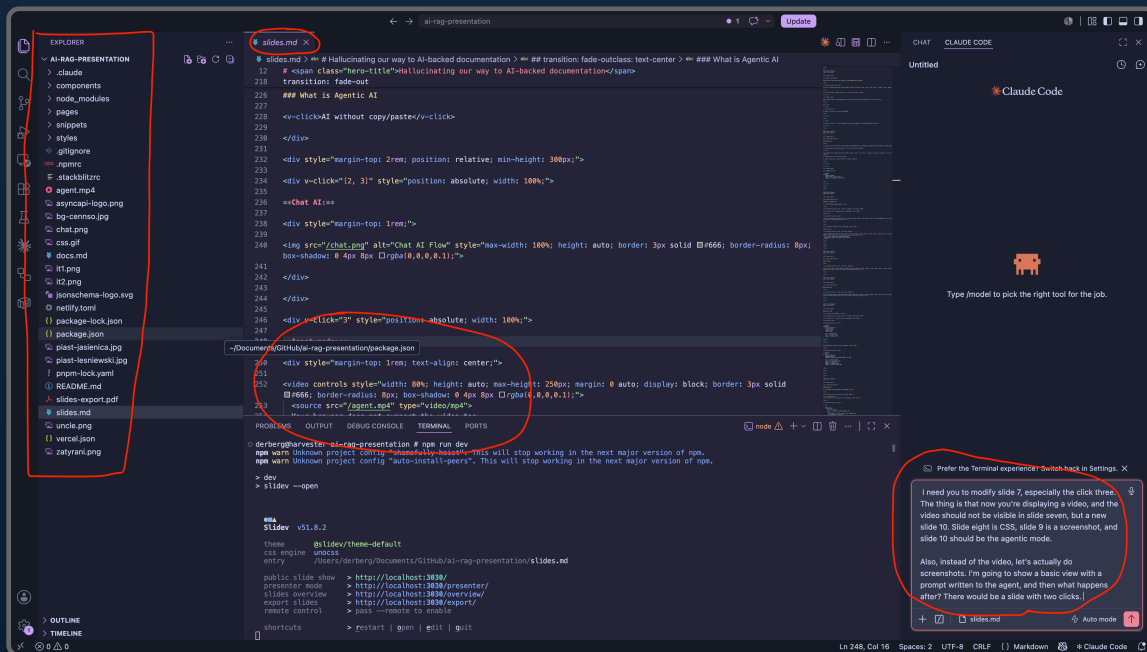
Agent
mode

Agent mode

AI without
copy/paste

Agent mode

AI without copy/paste



Agent mode

AI without copy/paste

The screenshot displays a VS Code editor interface with a presentation project. The Explorer sidebar on the left shows a file tree for 'AI-RAG-PRESENTATION' with files like 'components', 'node_modules', 'pages', 'snippets', 'styles', 'github.png', 'agent.mp4', 'stackblitzrc', 'asynapi-logo.png', 'bo-cards.jpg', 'chat.png', 'css.gif', 'docs.md', 'it.png', 'itd.png', 'jconschema-logo.svg', 'netlify.toml', 'package-lock.json', 'package.json', 'piant-javascript.png', 'piant-javascript.jpg', 'pnpim-lock.yaml', 'README.md', 'slides-export.pdf', 'slides.md', 'unide.png', 'vercel.json', and 'zatyran.png'. The main editor area shows the 'slides.md' file with a code editor view. The code is a Vite presentation script with a red circle highlighting the 'agent' component. The chat interface on the right shows a conversation with Claude Code, discussing the presentation slides and the 'agent' component. The chat messages include: 'I need you to modify slide 7, especially the click three. The thing is that now you're displaying a video, and the video should not be visible in slide seven, but in slide 10. Slide', 'Thought for 1s', 'Let me re-read that region to get the exact text.', 'Read slides.md (lines 278-300)', 'Thought for 1s', 'Edit slides.md', 'Added 48 lines', 'Done. Here's what changed in slides.md:', 'Slide 7 ("What is Agentic AI") -- removed the agent.mp4 video block (the old click 3). The Chat AI screenshot now just stays visible at click 2 instead of being replaced.', 'New slide 10 ("Agent mode") -- Inserted right after slide 9 (zatyran.pl screenshot) and before "What is MCP". It mirrors the original agent-mode styling with two clicks:', 'Click 1 -> "The prompt" -> /agent-prompt.png (basic view with the prompt written to the agent)', 'Click 2 -> "What happens after:" -> /agent-result.png (replaces the prompt screenshot)', 'Slide order is now: 7 Agentic AI -> 8 CSS -> 9 zatyran.pl screenshot -> 10 Agent mode -> 11 What is MCP.', 'Two things to note:', 'You still need to add the screenshot files agent-prompt.png and agent-result.png to the repo root (alongside chat.png, zatyran.png, etc). Rename them if you prefer different filenames and I'll update the refs.', 'agent-ask is no longer referenced anywhere -- you can delete it if you don't need it elsewhere.', 'Esc to focus or unfocus Claude', 'slides.md', 'Auto mode', 'Spaces: 2 UTF-8 CRLF', 'Markdown', 'Claude Code', 'Ln 241, Col 1'

Agent
mode

AI without
copy/paste

LESZY.RUN

STARTOFERTAWYDARZENIAKALENDARZ▼KONTAKT

ORGANIZUJESZ BIEG?

KALENDARZBIEGÓW

WSZYSTKIE WYDARZENIA W POLSCE

Setki biegów, marszów nordic walking i wydarzeń sportowych z całej Polski.

POMÓŻ ULEPSZYĆ

+ DODAJ WYDARZENIE

POLECAMY

za 117 dni

WILCZY PÓŁMARATON

17 października 2026 · Wilcza, Śląsk

TRAIL

Bieg 21 km

Canicross 21 km

Nordic Walking 11 km

SZCZEGÓŁY →

Szukaj po nazwie, miejscu...

BLISKO MNIE

LISTAKALENDARZMAPA

Typ: Wszystkie ▼

Region: Cała Polska ▼

Dystans: Wszystkie ▼

Cena: Wszystkie ▼

Kiedy: Najbliższe ▼

Znaleziono 895 wydarzeń

CZERWIEC 2026

26.06

BIEG DLA SŁONIA

Chorzów

ULICZNY

NOCNY

7 KM

ZGŁOŚ POPRAWKĘ

26.06

ROZTOCZAŃSKI FESTIWAL BIEGOWY- E...

Bosznia Dolna

ZAPISY ZAMKNIĘTE

TRAIL

NOCNY

NORDIC WALKING

ULTRAMARATON

7 KM / 5 KM / 60 KM / 42 KM +6

ZGŁOŚ POPRAWKĘ

26.06

BIEGAJ Z DUCHAMI - STRASZNA EXTREMEJURA 2026

Żarki

NOCNY

NORDIC WALKING

TRAIL

15 KM / 9 KM

ZGŁOŚ POPRAWKĘ

The big words

 **Agentic**

 **LLM**

 **RAG**

 **MCP**

 **Skills**

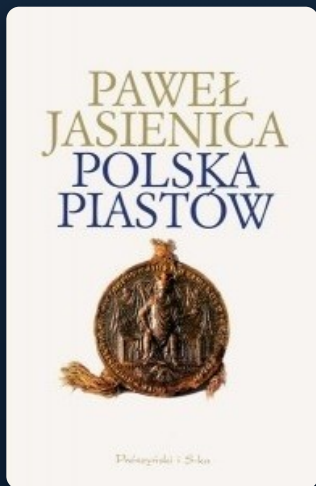
What is LLM

What is LLM



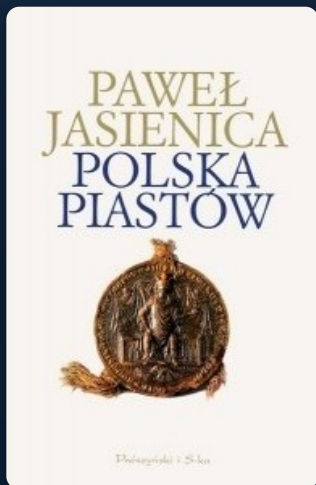
You don't know what the LLM knows about the Piast dynasty

You don't know what the LLM knows about the Piast dynasty



Polska Piastów – Paweł Jasienica

You don't know what the LLM knows about the Piast dynasty



Polska Piastów – Paweł Jasienica



Drapieżny ród Piastów – Sławomir Leśniewski

You don't know where the LLM gets its knowledge about AsyncAPI

You don't know where the LLM gets its knowledge about AsyncAPI

can I use protobuf with AsyncAPI

Yes, Protocol Buffers (Protobuf) can absolutely be used with AsyncAPI, though they serve different purposes:

- AsyncAPI describes your event-driven API (channels, messages, operations, servers, etc.).
- Protobuf defines the binary serialization format and message schema.

They complement each other well.

Option 1: Reference `.proto` schemas (recommended)

You can define your message in a `.proto` file:

```
<< proto
syntax = "proto3";
package user;

message UserCreated {
  string id = 1;
  string email = 2;
  string name = 3;
}
```

Then reference it from your AsyncAPI document:

```
<< YAML
asyncapi: 3.0.0

channels:
  user-created:
    address: user.created
    message:
      UserCreatedMessage:
        payload:
          schemaFormat: application/vnd.google.protobuf+version=3
          schema:
            $ref: "/schemas/user.proto/UserCreated"
```

The important field is:

```
<< YAML
schemaFormat: application/vnd.google.protobuf+version=3
```

You don't know where the LLM gets its knowledge about AsyncAPI

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  user-created:
    address: user.created
    message:
      UserCreatedMessage:
        payload:
          schemaFormat: application/vnd.google.protobuf;version=3
          schema:
            $ref: "../schemas/user.proto/UserCreated"
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The important field is:

```
<< YAML
schemaFormat: application/vnd.google.protobuf;version=3
```

AsyncAPI Local-First

Ask AsyncAPI

Answers about the AsyncAPI spec, concepts, tooling, and community — grounded in the official docs, with sources you can verify.

♥ AsyncAPI is built by people with passion, not large corporate funding. Support it

YOU

can I use protobuf with AsyncAPI

ASYNCAPI KNOWLEDGE

Yes, AsyncAPI can describe Protobuf payloads. Do this by using Protobuf as the payload schema format and embedding the Protobuf schema itself as a string.

- How it works:
 - In the payload, set payload.schemaFormat to a Protobuf media type (e.g., application/vnd.google.protobuf;version=3).
 - Put the actual Protobuf definition in payload.schema as a string.
 - Do not use \$ref to reference parts of the Protobuf schema; the Protobuf schema must be included directly as text.
- Example (YAML):

```
payload:
  schemaFormat: application/vnd.google.protobuf;version=3
  schema: "syntax = \"proto3\"; message User { string id = 1; string name = 2; }"
```

Sources:

- AsyncAPI define-payload — Schema formats: <https://www.asyncapi.com/docs/concepts/asyncapi-document/define-payload>

Blind spot

How can the LLM know about your closed-source product with docs behind a login page?

Blind spot

How can the LLM know about your closed-source product with docs behind a login page?



It can't – just a void

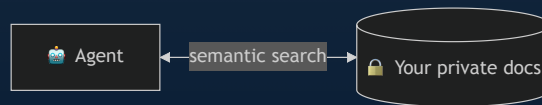
Blind spot

How can the LLM know about your closed-source product with docs behind a login page?



It can't – just a void

RAG



The big words

 **Agentic**

 **LLM**

 **RAG**

 **MCP**

 **Skills**

What is RAG

What is RAG

Semantic search over your own content

What is RAG

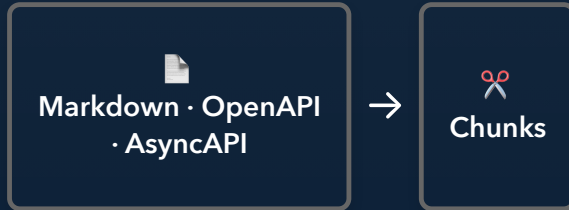
Semantic search over your own content



Markdown · OpenAPI
· AsyncAPI

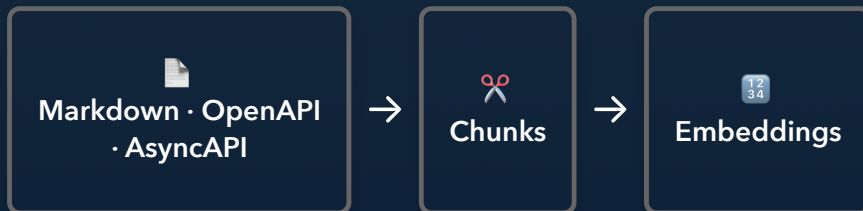
What is RAG

Semantic search over your own content



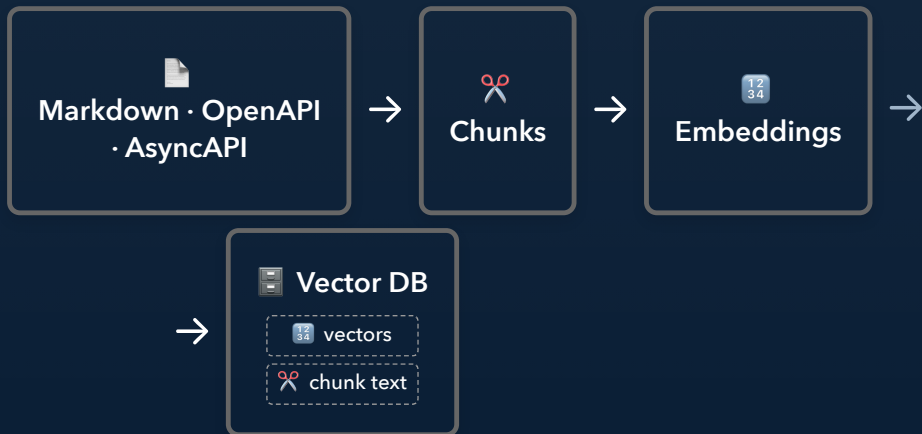
What is RAG

Semantic search over your own content



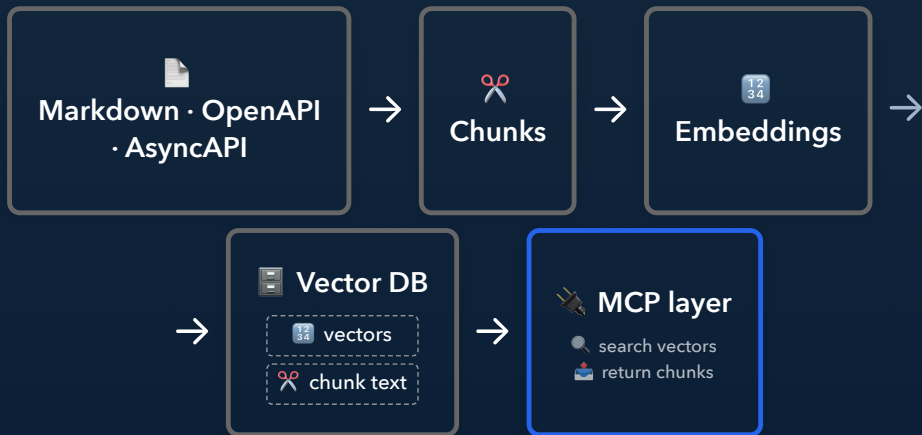
What is RAG

Semantic search over your own content



What is RAG

Semantic search over your own content



Chunks

Chunks

One doc → many structure-aware chunks

Chunks

One doc → many structure-aware chunks



Prose

```
{
  "chunk_id": "c4ce5efe...382289e",
  "chunk_type": "prose",
  "content": "## Multiple channels with single message when reply address is known\n\nThe request/reply pattern can also be implemented over multiple channels with a single message..",
  "line_start": null,
  "metadata": {
    "source_url": "https://www.asyncapi.com/docs/concepts/asyncapi-document/reply-info"
  },
  "source_file": ".opencrane/llmstxt/llms-full.txt",
  "source_name": "asyncapi-website",
  "token_count": 74
}
```


Chunks

One doc → many structure-aware chunks



Prose

```
{
  "chunk_id": "c4ce5efe_382289e",
  "chunk_type": "prose",
  "content": "## Multiple channels with single message when reply address is known\n\nThe request/reply pattern can also be implemented over multiple channels with a single message.",
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  "source_name": "asyncapi-website",
  "token_count": 74
}
```

Chunks

One doc → many structure-aware chunks

List item

```
{
  "chunk_id": "43ca503b_92cd7ef",
  "chunk_type": "list_item",
  "content": "1. The Generator receives the AsyncAPI Document as input.",
  "line_start": null,
  "metadata": {
    "breadcrumb_path": ".../asynccapi-document AsyncAPI document generation process",
    "depth": 0,
    "list_id": "7222577c705f7892",
    "list_style": "ordered",
    "parent_item_id": null,
    "position": 1,
    "sibling_ids": ["9714fd50_d554292", "b9728c45_fe3aca0", "0616eda6_59fbd81", "bc681ae6_eeee7a4", "537dc8cb_f176ba6"],
    "sibling_previews": ["2. The Generator sends to...", "3. The Parser validates t...", "4. If the Parser determin...", "5. At this point, the Gener...", "6. The Template Context i..."],
    "source_url": "https://www.asynccapi.com/docs/tools/generator/asynccapi-document",
    "total_siblings": 6
  },
  "source_file": ".opencrane/llmstxt/llms-full.txt",
  "source_name": "asynccapi-website",
  "token_count": 37
}
```

Chunks

One doc → many structure-aware chunks

List item

```
{
  "chunk_id": "43ca503b...92cd7ef",
  "chunk_type": "list_item",
  "content": "# ... AsyncAPI document generation process\n1. The **Generator** receives the **AsyncAPI Document** as input.",
  "line_start": null,
  "metadata": {
    "breadcrumb_path": ".../asyncapi-document AsyncAPI document generation process",
    "depth": 0,
    "list_id": "7222577c705f7892",
    "list_style": "ordered",
    "parent_item_id": null,
    "position": 1,
    "sibling_ids": ["9714fd50...d554292", "b9728c45...fe3aca0", "0616eda6...59fbd81", "bc681ae6...eeea7a4", "537dc8cb...f176ba6"],
    "sibling_previews": ["2. The **Generator** sends to...", "3. The **Parser** validates t...", "4. If the **Parser** determin...",
    "5. At this point, the **Gener...", "6. The **Template Context** i..."],
    "source_url": "https://www.asyncapi.com/docs/tools/generator/asyncapi-document",
    "total_siblings": 6
  },
  "source_file": ".opencrane/llmstxt/llms-full.txt",
  "source_name": "asyncapi-website",
  "token_count": 37
}
```

Chunks

One doc → many structure-aware chunks

JSON Schema

```
{
  "chunk_id": "1fcc25de...395bb4",
  "chunk_type": "json_schema",
  "content": {
    "description": "A unique id representing the application.",
    "format": "uri",
    "type": "string"
  },
  "line_start": null,
  "metadata": {
    "breadcrumb_path": "properties.id",
    "logical_parent": "root",
    "neighbor_chunks": ["bdf39a6c...294dec3", "6dad8d11...1c2c319", ...],
    "original_format": "yaml",
    "property_name": "id",
    "property_path": "id",
    "schema_element": "properties",
    "schema_title": "AsyncAPI 3.1.0 schema.",
    "schema_type": "json_schema",
    "schema_version": "http://json-schema.org/draft-07/schema",
    "source_url": "https://github.com/asynccapi/website/blob/master/config/3.1.0.json"
  },
  "source_file": ".opencrane/llmstxt/llms-full.txt",
  "source_name": "asynccapi-json-schema",
}
```

Chunks

One doc → many structure-aware chunks

JSON Schema

```
{
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  "chunk_type": "json_schema",
  "content": {
    "description": "A unique id representing the application.",
    "format": "uri",
    "type": "string"
  },
  "line_start": null,
  "metadata": {
    "breadcrumb_path": "properties.id",
    "logical_parent": "root",
    "neighbor_chunks": ["bdf39a6c...294dec3", "6dad8d11...1c2c319", ...],
    "original_format": "yaml",
    "property_name": "id",
    "property_path": "id",
    "schema_element": "properties",
    "schema_title": "AsyncAPI 3.1.0 schema.",
    "schema_type": "json_schema",
    "schema_version": "http://json-schema.org/draft-07/schema",
    "source_url": "https://github.com/asynccapi/website/blob/master/config/3.1.0.json"
  },
  "source_file": ".opencrane/llmstxt/llms-full.txt",
  "source_name": "asynccapi-json-schema",
}
```

The big words

 **Agentic**

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What is MCP

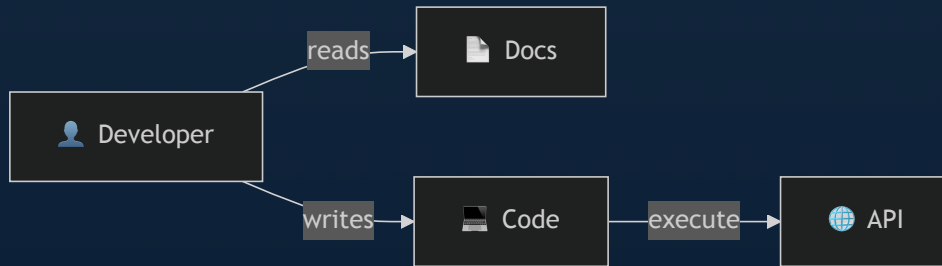
What is MCP

API for AI agents

What is MCP

API for AI agents

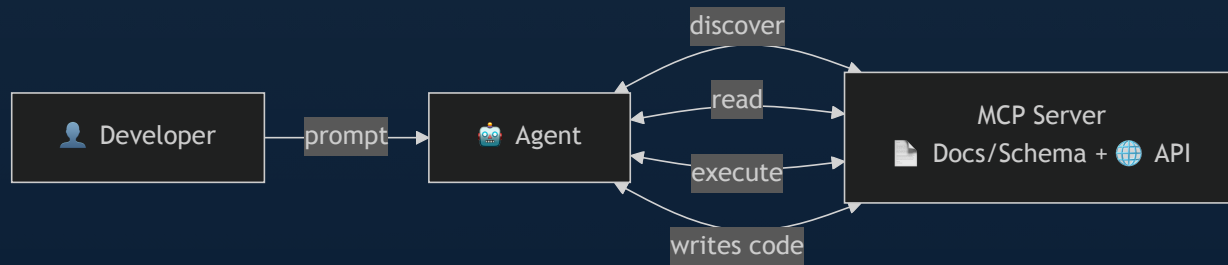
Traditional API:



What is MCP

API for AI agents

MCP:



What is MCP

API for AI agents

MCP:



Behind MCP

Behind MCP

What is MCP exposing to the agent

Behind MCP

What is MCP exposing to the agent

API

```
{
  "mcpServers": {
    "kubernetes": {
      "command": "npx",
      "args": ["mcp-server-kubernetes@3.2.0"],
      "env": {
        "K8S_SERVER": "https://prod-cluster.example.com",
        "K8S_NAMESPACE": "my-app",
        "ALLOW_ONLY_READONLY_TOOLS": "true"
      }
    }
  }
}
```

Behind MCP

What is MCP exposing to the agent

Custom Model:

```
{
  "mcpServers": {
    "3gpp-server": {
      "command": "npx",
      "args": ["3gpp-mcp-charging@3.0.2", "serve"]
    }
  }
}
```

Behind MCP

What is MCP exposing to the agent

RAG:

Local (uvx):

```
{
  "mcpServers": {
    "asynccapi-knowledge": {
      "command": "uvx",
      "args": ["asynccapi-knowledge-mcp=0.0.4"]
    }
  }
}
```

Hosted (HTTP):

```
{
  "mcpServers": {
    "asynccapi-knowledge": {
      "url": "https://derberg-asynccapi-knowledge-mcp.h
f.space/http"
    }
  }
}
```


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Skills

Skills

Knowledge & tools aren't enough – the agent needs your playbook

Skills

Knowledge & tools aren't enough – the agent needs your playbook

 **off the leash** → capable, but does whatever it wants